AIDAN SIRBU

E-mail: aidan.sirbu@mila.quebec | **Website:** aidansirbu.com
GitHub: github.com/asirbu77 | **LinkedIn:** linkedin.com/in/aidansirbu

EDUCATION

McGill University – Graduate Studies Montreal, QC
Ph.D., Computer Science (In progress) 2025-Present
Supervisor: Dr. Blake Richards

Western University – Bachelor's Studies London, ON
B.Sc. Honours (Western Scholar), Physics & Astronomy 2019-2023
Specialization: Medical Physics
Thesis: Quasi-Quantum Brain Energetics
Supervisor: Dr. Andrea Soddu
Graduated with Distinction

PUBLICATIONS & PREPRINTS

Sirbu, A., Zhang, C. Y., Yang, Y., Olveczky, B., Pereira, T., & Richards, B. (2026). A latent control model for realistic rodent motion. *Advances in Neural Information Processing Systems* (to appear). https://doi.org/10.48550/arXiv.2610.27279

Etcheverry, M., Miotti, P., **Sirbu, A.** et al. Reasoning with neural cellular automata. arXiv 2609.36126 (2026). https://doi.org/10.48550/arXiv.2609.36126

Drozdova, M.*, **Sirbu, A.***, Miotti, P. et al. Diffusion as a training curriculum for timestep-free iterative reasoning. NeurIPS BeNTo Workshop (2026). https://doi.org/10.48550/arXiv.2609.01449

Zhang, C. Y.*, Yang, Y.*, **Sirbu, A.*** *et al.* MIMIC-MJX: Neuromechanical emulation of animal behavior. *arXiv* 2511.20532 (2025). https://doi.org/10.48550/arXiv.2511.20532

Ellerman, F.*, **Sirbu, A.***, Brahms, A. *et al.* Spying on parahydrogen-induced polarization transfer using a half-tesla benchtop MRI and hyperpolarized imaging enabled by automation. *Nat Commun* **14**, 4774 (2023). https://doi.org/10.1038/s41467-023-40539-9

***Co-first contribution**

HONOURS & SCHOLARSHIPS

Graduate Excellence Award – McGill University ($6,000)	2026-2027
Canada Graduate Research Scholarship-Master's – NSERC ($27,000)	2026-2027
Excellence Scholarship, MSc – UNIQUE NeuroAI Institute ($10,000)	2026-2027
Master's Research Scholarship – Nature and Technologies ($20,000 per year)	2025-2026

J.B Bancroft Science Prize and Medal ($250), Western University 2023

Western Gold Medal Honours Specialization in Medical Physics, Western University 2023

Gonfalon Bearer for the Faculty of Science's Graduating Class, Western University 2023

Undergraduate Student Research Award ($6,000), NSERC 2023

S.R. Valluri Scholarship in Mathematical or Theoretical Physics ($1,000), Western Uni. 2022

Rise-Globalink Research Award ($6,000), Mitacs & DAAD 2022

Mary Ann Underwood Small Global Opportunities Award ($2,000), Western University 2022

International Learning Award ($1,000), Western University 2022

In-Course Scholarship Years III & IV ($1,400), Western University 2021/2022

Undergraduate Student Research Award ($6,000), NSERC 2021

John Gordon McIntosh Scholarship for Second Year Physics ($500), Western University 2021

Dean's Honour List for four consecutive years, Western University 2019-2023

The Western Scholarship of Excellence ($2,000), Western University 2019

INVITED TALKS

Embodied Agents for NeuroAI: Building brain inspired foundation models for motor control.
ARNI Annual Retreat 2025, Columbia University, New York City, NY. Nov 2025

OTHER PEER-REVIEWED CONTRIBUTIONS

MIMIC-MJX: Neuromechanical Emulation of Animal Behavior. Charles Y. Zhang, Yuanjia Yang, **Aidan Sirbu**, et al. COSYNE 2026 (Poster).

RESEARCH EXPERIENCE

Google Montreal, QC

Student Researcher, Paradigms of Intelligence Team June 2026 – Present

Research streams:

- Looped iterative reasoning
- Neural cellular automata
- Hierarchical reinforcement learning for embodied systems

Western University London, ON

Research Assistant, Physics & Astronomy May 2023 – Aug 2023

Principle Investigator: Dr. Andrea Soddu

- Formulated my own independent research project.
- Designed and implemented large, scalable spiking neural networks based on Izhikevich's spiking neuron model in Python from scratch.

- Developed a learning rule specifically designed to promote pattern encoding in spiking networks.
- Built a foundational understanding of artificial neural networks.

Universitätsklinikum Schleswig-Holstein & Kiel University Kiel, Germany
Research Assistant, Section Biomedical Imaging May 2022 – Aug 2022
Principle Investigator: Dr. Jan-Bernd Hövener

- Designed, built, and ran a portable, fully automated parahydrogen-induced polarizer.
- Leveraged skills in MATLAB to interface with and program pulse sequences in a 0.5T tabletop MRI machine.
- Developed data analysis tools for monitoring the evolution of spin order distribution in real-time.
- Co-authored a manuscript and familiarized myself with the process of peer-revision.

Western University London, ON
Research Assistant, Physics & Astronomy May 2021 – April 2022
Principle Investigator: Dr. Alexei Ouriadov

- Prototyped and ran a stopped-flow xenon polarizer.
- Measured Xe polarization using a low-field pre-clinical MRI.
- Performed IDL and MATLAB based k-space simulations.

Academic Volunteering

Mila Delegation @ NeurIPS 2025 2025
Delegation Member, Mila - Quebec AI Institute

Physics Undergraduate Conference Committee (PhUnC) 2022-2023
Presentation Coordinator, Western University

Physics Undergraduate Conference Committee (PhUnC) 2021-2022
Outreach, Western University